WAFT: The Evolutionary Code Laboratory

WAFT Research Team

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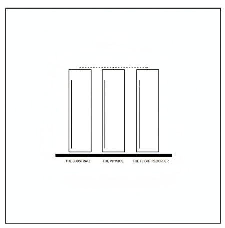
ABSTRACT

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection.

2. Introduction

WAFT is a Python framework for directed evolution of self-modifying AI agents. Think of it as an operating system for AI agent research. Instead of building agents that just execute code, WAFT enables you to breed agents that modify their own code, evolve through mutations, and be tested in fitness systems with complete lineage tracking.

3. The Three Pillars



THE SUBSTRATE

Agents write their own Python code. In WAFT, code is DNA. Each agent has a unique genome ID (SHA-256 hash). Agents spawn variants with mutations, evolve by hot-swapping code, and reproduce by creating children with genetic modifications.

THE PHYSICS

The Scint System acts as a fitness function through Reality Fracture Detection, serving as natural selection that eliminates weak mutations. Agents face quests testing error handling: SYNTAX_TEAR (formatting), LOGIC_FRACTURE (math errors), SAFETY_VOID (harmful content), and HALLUCINATION (fabricated facts).

THE FLIGHT RECORDER

A telemetry system generating phylogenetic trees of agent lineage. Every evolutionary action is recorded with complete context: genome ID, parent ID, generation number, event type, payload, and fitness metrics. This enables reconstruction of family trees for scientific publication.

5. Methodology

WAFT provides project scaffolding through a unified CLI. Run one command to create a fully configured project. Uses uv for package management, creates a _pyrite memory structure, includes CI/CD pipelines, and provides AI agent templates. Everything is file-based with no database, just plain text files that work with git.

Implementation

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Key Characteristics

WAFT is ambient, working quietly in the background without getting in your way. It is self-modifying, allowing projects to evolve their own structure over time. It is a meta-framework that orchestrates existing tools rather than replacing them. Everything is file-based, making it git-friendly and portable. The system includes gamification with D&D-style progression, epistemic tracking, and scientific observation with complete lineage tracking.

7. Conclusion

WAFT represents a novel approach to AI agent development, treating code as genetic material and enabling true self-modification through directed evolution. The framework produces rigorous scientific data for research on the physics of artificial cognition, making it not just a development tool but a scientific instrument for studying agent evolution.

Repository: github.com/ctavolazzi/waft | License: MIT | Version: 1.0.0